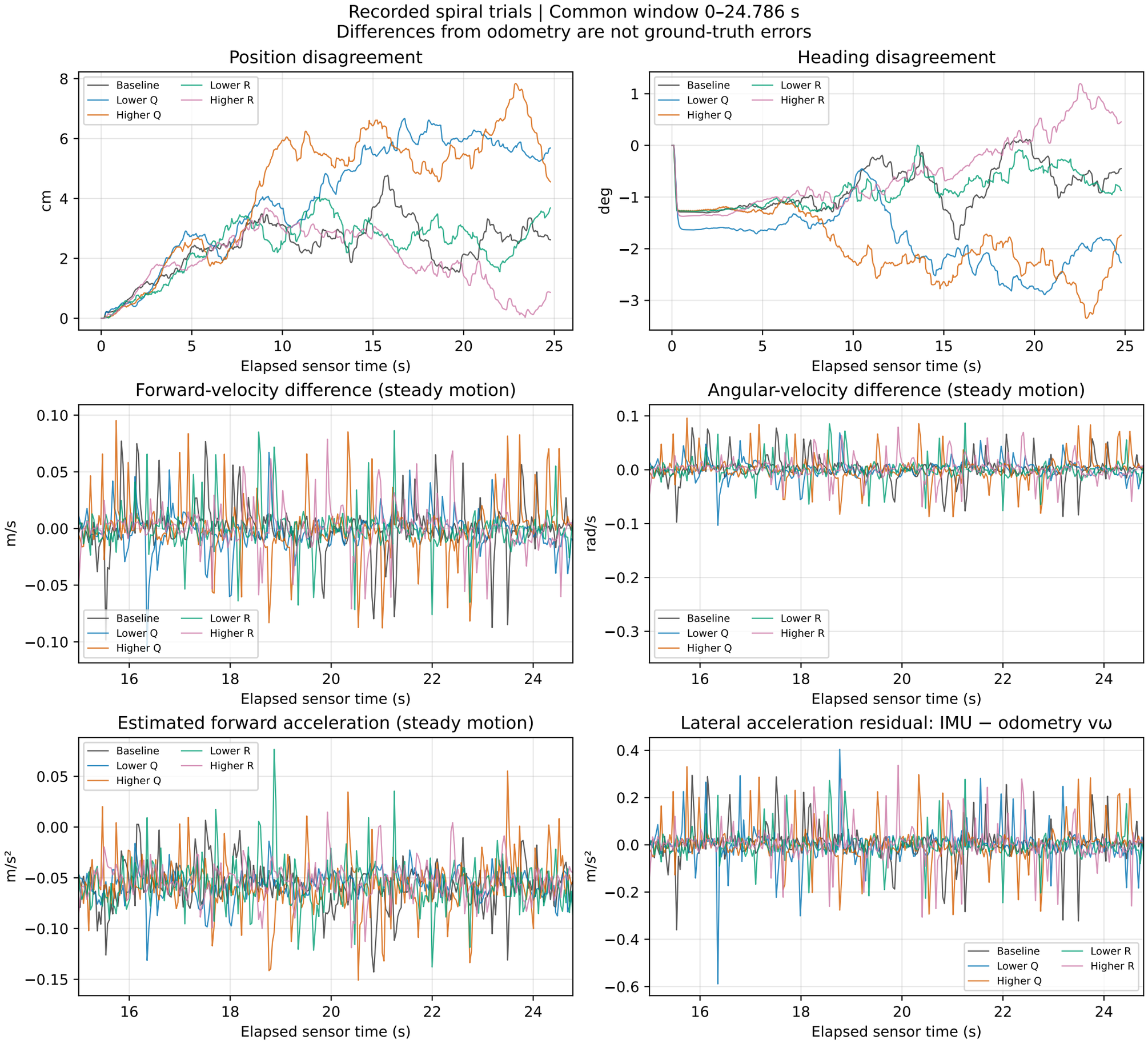

Recorded spiral trials | Common window 0–24.786 s
Differences from odometry are not ground-truth errors
Position disagreement
cm
Elapsed sensor time (s)
Heading disagreement
deg
Elapsed sensor time (s)
Forward-velocity difference (steady motion)
m/s
Elapsed sensor time (s)
Angular-velocity difference (steady motion)
rad/s
Elapsed sensor time (s)
Estimated forward acceleration (steady motion)
m/s²
Elapsed sensor time (s)
Lateral acceleration residual: IMU − odometry vω
m/s²
Elapsed sensor time (s)
Baseline
Lower Q
Higher Q
Lower R
Higher R